# Session 7 (PROTECT & WRITE: ethics, integrity, Methods and Results): Exercise sheet

Foundations of Clinical Research · Level 1 · Neudata · *#ClearDataClearImpact* · Week 7 · Saturday 12 December 2026 · online, 19:30–22:30

**Name:** _____ **Department:** _____

Type your answers into this Word file or print the PDF.

| Part | Exercise | When | How |
|---|---|---|---|
| A | Integrity scenarios: green, amber or red? | 20:25 (5 min) | Whole class, polls: 3 scenarios in class, 3 at home |
| B | Proposal builder 7: ethics, registration, feasibility, publication plan | 20:30 (15 min) | The instructor models it on the running example; you write your own Part 4 at home |
| C | **Manuscript workshop 1**: the Results section for your FLUID-ICU question | 21:35 (45 min) | 30 min whole class on Q1, then 15 min on your own question. Finish at home |

Files you need (sent by email): the results pack for your question (`Results_Pack_QN.pdf`), the manuscript template (`Manuscript_Template_IMRaD.docx`), Paper 1 (Boulet 2024) and Paper 2 (Hyun 2023).

## Part A: Integrity scenarios: green, amber or red? (5 minutes, 20:25)

In class we vote on scenarios 1, 3 and 4 in a poll; scenarios 2, 5 and 6 are homework. For each scenario choose:

- **GREEN**: acceptable as described.
- **AMBER**: acceptable **only if** something is fixed or disclosed (say what).
- **RED**: research misconduct or a serious breach. Do not do it.

Write one sentence explaining your colour.

| # | Scenario | Colour | Why? |
|---|---|---|---|
| 1 | Three patients' day-3 fluid charts are lost. A resident "remembers roughly" what their balances were and enters values so the dataset is complete. | | |
| 2 | After seeing the results, the team removes two patients with "strange" fluid values because they weaken the association. The paper does not mention it. | | |
| 3 | The head of department did not take part in the study but asks to be last author "because it is my department". | | |
| 4 | (a) A nurse uses an AI chatbot to improve the English of the Discussion she wrote, checks every sentence, and states this in the Acknowledgements. (b) A colleague uploads the study spreadsheet (names and hospital numbers included) to a free public chatbot and asks it to "run the statistics". | (a) | |
| | | (b) | |
| 5 | A cohort of patients with sepsis like FLUID-ICU is published as three papers (mortality, acute kidney injury and ICU stay) in three journals, none mentioning the others. | | |
| 6 | Two paragraphs of a published review are copied word for word into your Introduction. The review is cited at the end of the paragraph, but there are no quotation marks. | | |

## Part B: Proposal builder 7: ethics, registration, feasibility and publication plan

**In class (20:30, 15 minutes):** the instructor fills in this worksheet live on the shared screen for the running example (a retrospective cohort of fluid balance and 28-day mortality in adults with septic shock in our ICU). You suggest answers in the chat.

**At home this week:** fill it in for **your own** proposal. It becomes Part 4 of your written proposal and slides 5 (ethics and feasibility) and 6 (publication plan) of your presentation. Everyone submits the written proposal and the 6 slides by email before 19:30 on Saturday 19 December 2026, Session 8 (`Proposal_Brief.pdf`).

## B1. Ethics and consent

| Question | Your answer |
|---|---|
| Which ethics committee will review the study? What type of review do you expect (exempt / expedited / full)? The committee decides. | |
| Who are the participants, and can they decide for themselves when they become eligible? | |
| Who gives consent (patient / legal representative / deferred consent / waiver requested)? Use the ICU consent flowchart (Session 7 slides). | |
| Who takes consent, when and where? | |
| What are the risks and burdens for participants, and how are they minimised? | |
| Does the study need registration (trials: yes, before the first patient, in ClinicalTrials.gov or another WHO registry; systematic reviews: PROSPERO; observational: recommended)? | |

## B2. Confidentiality

| Question | Your answer |
|---|---|
| Which direct identifiers will you collect (if any), and why? | |
| How will each patient get a study code? | |
| Where is the key file (code to identity) kept, and who can open it? | |
| Where will the de-identified data be stored (institution-approved, password-protected)? | |

## B3. Feasibility check (tick and comment)

| ✓ | Question | Comment |
|---|---|---|
| ☐ | **Time:** realistic timeline (ethics 1–3 months, recruitment, analysis, writing)? | |
| ☐ | **Money:** budget headings named (staff time, materials/tests, publication fee)? | |
| ☐ | **People:** who collects data (nights, weekends), who analyses, protected time? | |
| ☐ | **Patients:** how many eligible patients did your unit see last year? How many can you realistically enrol? | |
| ☐ | **Data:** are the key variables recorded routinely? How much is likely missing? | |

Biggest feasibility risk: _____ One fix: _____

## B4. Publication plan (slide 6 of your presentation)

| Item | Your answer |
|---|---|
| Working title (from your PICO: design, I vs C, P, O; 10–16 words) | |
| Target journal (and a back-up), checked in DOAJ / NLM Catalog, with fees and licence known | |
| Reporting guideline (CONSORT / STROBE / STARD / PRISMA / SPIRIT / CARE) | |
| Registration needed? Where? (from B1) | |
| Authors (all four ICMJE criteria) and their roles (CRediT) | |
| Conflicts of interest and funding to declare | |
| Data-sharing statement (what, where, under which conditions) | |
| Conference where you would present an abstract first | |

**Structured-abstract skeleton.** One line each. Results are the results you *expect to report* (format, not numbers).

| Section | One line |
|---|---|
| Background | |
| Methods | |
| Results (expected) | e.g. "28-day mortality: n/N per group, RR (95% CI)" |
| Conclusion | Write the sentence you would use if the result is positive, and the one you would use if it is not. |

# Part C: Manuscript workshop 1: the Results section (45 minutes, 21:35)

You write the **Results section** for your FLUID-ICU question, using only your results pack. Only the Results section is required. You may add other sections in the same file (`Manuscript_Brief.pdf`).

| Question | Topic | Results pack |
|---|---|---|
| Q1 | Positive fluid balance by day 3 and 28-day mortality | `Results_Pack_Q1.pdf` |
| Q2 | Cumulative fluid balance and acute kidney injury (AKI) | `Results_Pack_Q2.pdf` |
| Q3 | Cumulative fluid balance and ventilator-free days | `Results_Pack_Q3.pdf` |
| Q4 | Initial lactate and 28-day mortality, complete case vs multiple imputation | `Results_Pack_Q4.pdf` |
| Q5 | Fluid balance and ICU length of stay among survivors | `Results_Pack_Q5.pdf` |

## How the 45 minutes are used

| Clock | Who | Task |
|---|---|---|
| 21:35–21:45 | Whole class, Q1 | Participants and the Table 1 sentence. You suggest sentences in the chat. The instructor types the best one |
| 21:45–21:57 | Whole class, Q1 | Deaths per group, crude and adjusted OR with 95% CI |
| 21:57–22:02 | Whole class, Q1 | Other analyses: Cox model and multiple imputation |
| 22:02–22:05 | Whole class | A volunteer checks every number against the pack |
| 22:05–22:20 | You | Start the Results for YOUR question, following the class draft |

The class draft is sent as a PDF after the session. If you chose Q1, write it in your own words.

## Results checklist (STROBE items)

- ☐ Participants first: numbers enrolled, analysed and excluded, with reasons (13)
- ☐ Table 1 in one or two sentences, with numbers. Missing data reported (14)
- ☐ Outcome numbers per group, with denominators (15)
- ☐ Main result: **crude and adjusted** estimate with 95% CI (16)
- ☐ Other analyses in your pack: sensitivity, imputation or subgroup (17)
- ☐ Table 1 and one main table or figure that stand alone
- ☐ No interpretation ("because", "suggests") and no causal words
- ☐ A working title and the AI-use statement

## Question-specific reminders

- **Q3:** ventilator-free days are skewed and count death as zero.
- **Q4:** report both the complete-case and the multiple-imputation estimates.
- **Q5:** survivors only. Use medians and IQR.

## Methods (taught in 7.3, optional to submit)

If you also submit the Methods, use STROBE items 4 to 12: design and setting, participants, variables, data sources, bias, study size ("all eligible patients in the cohort were included"), statistical methods and missing data. The facts are in your results pack, `SAP_FLUID-ICU.pdf` and `Data_Dictionary.pdf`.

## Deliverable

Your Results section, named `Surname_Firstname_Results_QN.docx`, emailed to your reviewer and to the instructor (myluong1710@gmail.com) **by 23:59 on Sunday 20 December 2026 (Vietnam time)**. Your reviewer is announced at the end of today's session.